

RENANCE

FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN

PHY 212 — Past Questions & Materials
by Renance

Course: PHY 212
Institution: Federal University of Petroleum Resources, Effurun
Papers in this pack: 4
Levels: 200, 300

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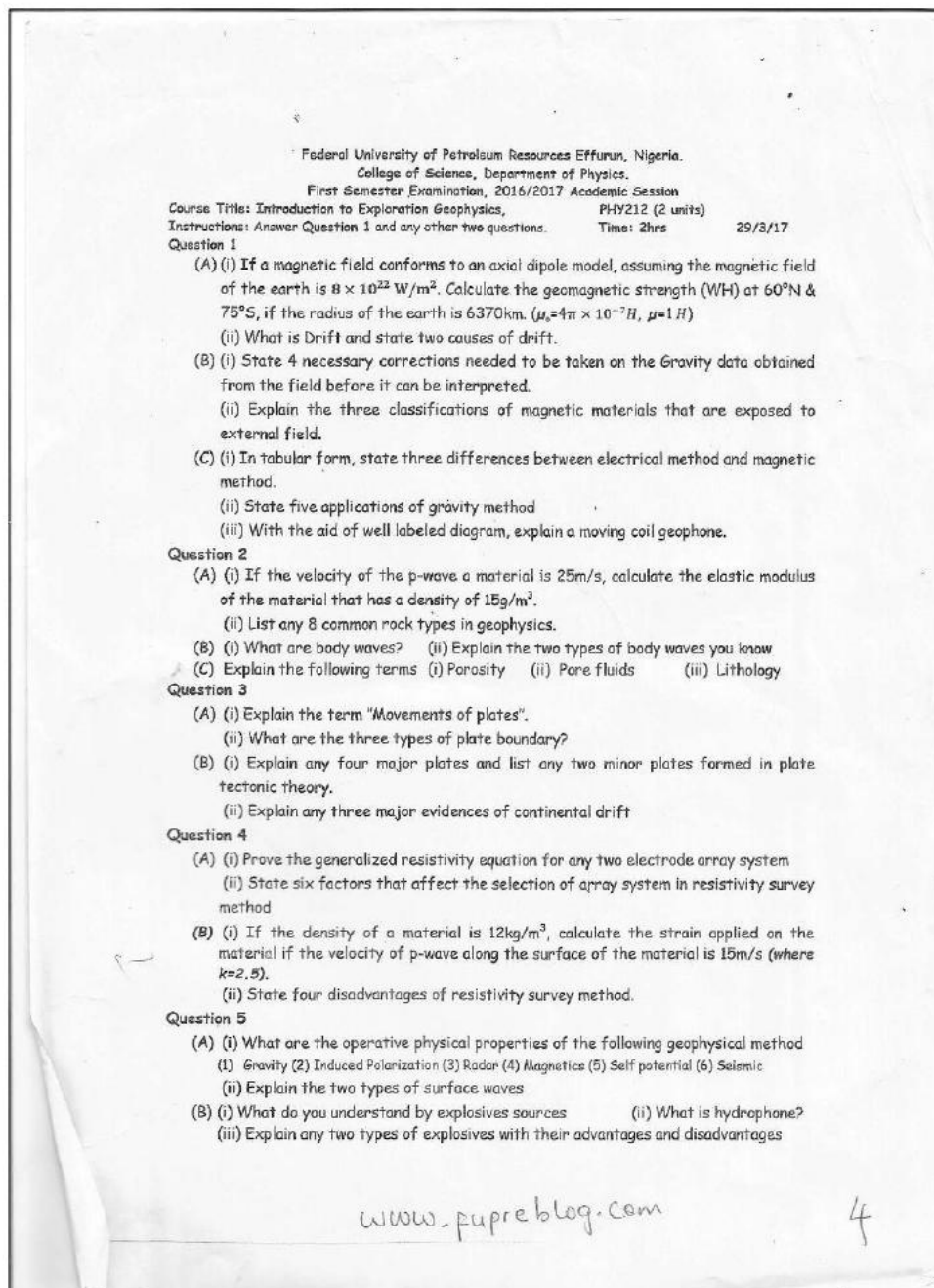
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Paper 1 of 4 — PHY 212 - 2016/2017 - First Semester - Introduction to Exploration Geophysics

Level 300 • First Semester • PastQuestion

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Paper 2 of 4 — PHY 212 - 2014/2015 - First Semester - INTRODUCTION TO EXPLORATION GEOPHYSICS

Level 200 • First Semester • PastQuestion

FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN

COLLEGE OF SCIENCE

DEPARTMENT OF PHYSICS

ST

1 SEMESTER EXAMINATION (2014/2015 SESSION)

PHY 212 (INTRODUCTION TO EXPLORATION GEOPHYSICS) (2 Units)

Instruction: Answer any three questions Time: 1hr 30mins

1. (a) Explain two major characteristics of geophysical methods and give two examples of each.
(b) Explain the problem with ambiguity in geophysical interpretation.
(c) List for geophysical techniques and their measured parameters.
2. (a) Explain six ways at which Rocks and Minerals in the earth vary.
(b) State three advantages of the gravity method of geophysical techniques.
3. (a) Briefly explain Seismic geophysical method.
(b) In a tabular form, state four advantages and disadvantages of active and passive geophysical methods.
4. (a) There are two major classes of Seismic Waves Explain?
(b) Explain (i) Seismic reflection
(ii) Seismic refraction
You can support your explanation with a diagram.
5. (a) Explain briefly Magnetic geophysical method.
(b) State the main difference between the resistivity (including induced polarization) methods and electromagnetic method.

Paper 3 of 4 — PHY 212 (INTRODUCTION TO EXPLORATION GEOPHYSICS)

Level 200 • First Semester • PastQuestion

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Paper 4 of 4 — PHY 212 2016 2017 EXAM

Level 200 • First Semester • PastQuestion

Federal University of Petroleum Resources Effurun, Nigeria.

College of Science, Department of Physics.

First Semester Examination, 2016/2017 Academic Session

Course Title: Introduction to Exploration Geophysics, PHY212 (2 units)

Instructions: Answer Question 1 and any other two questions. Time: 2hrs 29/3/17

Question 1

(A) (i) If a magnetic field conforms to an axial dipole model, assuming the magnetic field of the earth is $8 \times 10^{22} \text{ W/m}^2$. Calculate the geomagnetic strength (WH) at 60°N & 75°S , if the radius of the earth is 6370km. ($u=40 \times 10^{-7}\text{H}$, $w=1\text{H}$)

(ii) What is Drift and state two causes of drift.

(8) (i) State 4 necessary corrections needed to be taken on the Gravity data obtained from the field before it can be interpreted.

(ii) Explain the three classifications of magnetic materials that are exposed to external field.

(C) (i) In tabular form, state three differences between electrical method and magnetic method.

(ii) State five applications of gravity method ,

(iii) With the aid of well labeled diagram, explain a moving coil geophone.

Question 2

(A) (i) If the velocity of the p-wave a material is 25m/s, calculate the elastic modulus of the material that has a density of 15g/m^3 .

(ii) List any 8 common rock types in geophysics.

(B) (i) What are body waves? (ii) Explain the two types of body waves you know

(C) Explain the following terms (i) Porosity (ii) Pore fluids (iii) Lithology

Question 3

(A) (i) Explain the term "Movements of plates".

(ii) What are the three types of plate boundary?

(B) (i) Explain any four major plates and list any two minor plates formed in plate tectonic theory.

(ii) Explain any three major evidences of continental drift

Question 4

(A) (i) Prove the generalized resistivity equation for any two electrode array system

(ii) State six factors that affect the selection of array system in resistivity survey method

(B) (i) If the density of a material is 12kg/m^3 , calculate the strain applied on the EI material if the velocity of p-wave along the surface of the material is 15m/s (where

$k=2.5$).

(ii) State four disadvantages of resistivity survey method.

Question 5

(A) (i) What are the operative physical properties of the following geophysical method

(1) Gravity (2) Induced Polarization (3) Radar (4) Magnetism (5) Self potential (6) Seismic

(ii) Explain the two types of surface waves

(8) (i) What do you understand by explosives sources (ii) What is hydrophone?

(iii) Explain any two types of explosives with their advantages and disadvantages

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